

Why Chess Players Should Learn Computer Security

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ABSTRACT

This work shows the latest developments on chess frauds, their connections with security in computer science and how they can be fought. With the current organization of chess tournament, it is almost impossible to prevent third parties to interfere in a game or to guarantee the competition fairness. The primary concerns of a chess arbiter are to forbid players from receiving external advices, and coalition of players from biasing the outcome of the tournament. We are facing the same problem as those encountered when moving from human to human authentication to remote authentication of humans or machines. In this work, we demonstrate that chess frauds are easy to mount due to covert or subliminal channels. We describe chess frauds and link them to their analogues in the computer world. Based on these transpositions, we advocate for two radical transformations to ensure the fairness of chess tournaments.

Keywords

Chess, authentication, cheating, covert channels, chess grandmaster problem, prisoner problem.

1. INTRODUCTION

Chess still fascinates generations of computer scientists. Many great researchers such as von Neumann [47], Shannon [41], or Turing to name a few have spent time on studying chess programming [36]. In security, it has also inspired several works. Conway [13] with the *chess grandmaster problem* gave the first occurrence in the literature of man-in-the-middle attacks. His work was pursued and extended to authentica-

tion protocols by Desmedt, Goutier and Bengio [15] with the *mafia fraud* and later by Beth and Desmedt in [4] with the *terrorist fraud*.

The origin of this paper comes from recent news about cheating in chess tournaments. Indeed, an interesting cheating allegation occurred at the 2006 World Open in Philadelphia [10, 30]. After an impressive streak of wins, an unknown player was caught bearing several electronic devices including a cover earring device [37]. He was accused of getting help from an accomplice with this device. In the same year, Topalov accused Kramnik of cheating during their match for the world title at Elista [11]. Topalov's accusations against Kramnik were motivated by his opponent suspicious behavior: he visited the bathroom with an unusual frequency. Topalov's supporters claimed that Kramnik received external help in the bathroom (which were not under camera surveillance). They are many other examples of players being caught using wireless devices but it is not the only solution for cheating. Recently, several French grandmasters were accused by the French federation of fraud during the chess Olympiad 2010 at Khanty-Mansiysk [17, 32]. It seems that they agreed to help one of their fellow to win his games during the Olympiad. They used a combination of cell-phones, computer, and body language to inform the grandmaster of the best moves to play. A famous story of collusion allegations also occurred at the 5th Candidates Tournament at Curaçao in 1962 [46]. These allegations have motivated the choice of the Fédération Internationale Des Échecs (FIDE) for changing the organization of the battle for the world title. Players collusion is a critical issue for the fairness of chess tournaments. The question following these two examples is whether the next World Chess Champion will be a chess player or a script kiddie. Indeed, with the advances in new technologies, anybody has access to a computer capable of facing most of professional chess players. Combining it with miniaturized communication devices, our script kiddie is quite close to become the new Kasparov.

Hence, we are interested in two types of frauds in chess

games and tournaments, and we make the links between these chess frauds and the frauds that are well-known in computer security. First, the players can receive covert help from the public during a game. The arbiter wants to prevent such exchanges of information that we denote *external advisors*. Second, groups of players can misbehave during a tournament to increase the probability that one of them win. The arbiter needs also to deal with any form of *players collusion*. These attacks, *i.e.*, external advisors and collusion, are a major concern for nowadays chess tournament organizers and arbiters. We give several examples of fraud allegation to show the reality of those threats and the difficulties of arbiter's life.

To understand external advisors and players collusion, results and concepts from computer security are of great help. Covert advices are a special case of *covert channels*. The difficulty of setting up covert channels in our context is that we are not using computers but humans. Instead of reading files and measuring time, we are using the five human senses and our brain. A first challenge is to find a concise representation of the transmitted information. This can be solved using information theory and algorithms that are not too intensive for the human brain. With such covert channels, players and accomplices communicate without being detected by an arbiter. The accomplice just needs to get access to a computer running a chess software [35] to provide useful advices to the player. This connection is made easier with wireless technologies.

To deal with collusion frauds, we must render harmless any agreement between the malicious players prior to the tournament. Moreover, there must be no communications between the players during the different rounds of the tournament. We reach the problem of subliminal channels as in the *prisoner problem* of Simmons [42, 43, 44]. Isolation/confinement [29] are mandatory to defeat these frauds. They are enforced in what we define as a Jail tournament. Players are put into cells in which they cannot communicate with the outside or any other players. To prevent any agreement, a player does not know the identity of its adversary. This information is only available to the arbiter who has a full control on the games course. Finally to prevent communication through the chessboard channel, our recommendation is to use Fischer random chess (FRC) instead of classic chess.

The contribution of this paper is threefold: (a) the analysis of frauds recently discovered in chess tournaments and their connections with computer security, (b) the design of schemes to transmit concisely chess moves in order to mount frauds, (c) the proposal of solutions to prevent chess frauds.

The rest of the document is organized as follows. The link between chess rules and isolation in access control scheme is highlighted in the following section. Sections 3 and 4 make up the *Chess Cheating Cookbook*, *i.e.*, all the things you must know if you want to cheat at chess. Section 3 describes different classes of frauds and how they are related to computer security. It includes the *chess grandmaster problem*, external advisors and players collusion. In these previous attacks, covert channels play a central role. The complexity of creating covert channels is discussed in Section 4. Using informa-

tion theory, we analyze how a move can be coded concisely in order to be easily exchanged between the cheaters. Finally, Section 5 takes the view point of the arbiter by describing Jail tournament and Fischer random chess jail tournament.

2. RULES

In this section, we recall the chess rules that matter for our security problem. The rules of chess are all codified in the FIDE Laws of Chess [19]. A chess game involves two players, an 8×8 board and thirty two pieces (sixteen white and sixteen black). A clock is used to control and limit the time spent by both players. The result of a game is a win if a player is checkmate or if somebody resigns. It is a drawn if there is a stalemate, or more than 50 moves without any pawn moves are made, or both players agree on it. We do not give more details on the conduct of chess game. Such information can be found in numerous books (see [8, 9, 22]).

Chess players are social: they gather in tournament to find who is the strongest player at the moment. A tournament consists of a finite number of rounds n . At a each round, an arbiter defines the pairing: who is playing against who. There exist several pairing possibilities like Round-Robin, Swiss-system. . . . After the last round, the result of the tournament is proclaimed. The reward for the winners can be a trophy, the access to another chess event, money. . . . This reward can motivate the misconduct of certain players.

According to Articles 12 and 13 of [19] (see Appendix A), the arbiter must guarantee that the moves are the results of the own thinking of the players and that no third party is interfering. Two strategies can be used to insure fairness. First, the players produce a proof for each move made showing that they are their own: authentication of the players' moves. Second, the arbiter guarantees that the players cannot communicate with anybody else. This is an isolation strategy. The authentication strategy is particularly difficult to implement in practice, then we focus on the isolation strategy. Transposed in the language of a computer scientist, the arbiter has to solve *the problem of confining the public during a chess game so that they cannot transmit information to any of the players*. The previous statement is in fact very close to the own words of Lampson [29]. Controlling the information exchanged by the users of a system has been studied for years in computer science.

3. ATTACKS AGAINST CHESS GAME

The latest work dealing with chess and security is the analysis of the Internet Chess Club by Black, Cochran, and Gardner [5, 6] (online chess gaming). The connection between chess and security has been used a few time to support the understanding of several key-concepts in protocol designs. This section shows how chess cheating is linked to computer security [4, 15].

In our analysis, we consider a chess tournament in which we have honest and dishonest participants. We always assume that the arbiter is trusted. We do not discuss the case of online or postal chess (except by remembering the chess grandmaster problem and in Section 5).

3.1 Chess grandmaster problem

In 1976, Conway [13] described a first type of chess cheating known as the chess grandmaster problem. A little girl Anne-Louise, who has never heard of chess, wants to face two grandmasters, Fischer and Spassky in correspondence chess games. Has she lost her mind? No, she decides to play the black side against Fischer and the white side against Spassky (see Figure ??).

In this configuration, Fischer sends his first move to Anne-Louise. She copies this move and sends it to Spassky. Then, she waits Spassky's move and forwards it to Fischer. And so on, until she either wins one of the games, or draws both of them. Anne-Louise only relays the moves between the two grandmasters. The two chess grandmasters think they are actually playing against a little girl, while they are only playing against each other.

LINKS WITH COMPUTER SECURITY. This problem leads to the *man in the middle* attack used to defeat key exchange and authentication protocols. Desmedt, Goutier, and Bengio [15] transposed this fraud against (Feige-)Fiat-Shamir protocol [20, 21]. It was called the *mafia fraud*, as a reference to the famous Shamir's claim: "I can go to a mafia owned store a million times and they will not be able to misrepresent themselves as me." However, Desmedt *et al*, proved that Shamir was wrong. Indeed, an adversary, without knowing anything about the authentication protocol, nor breaking the cryptographic primitives can defeat an authentication scheme such as challenge/response. She just needs to relay the verifier challenges to any legitimate prover, and to forward the prover's answers to the verifier. In that way, the adversary can impersonate the prover to the verifier while the honest parties think they communicate to each other. This attack has been applied time to time in various domains: smartcard [16], RFID [25], car authentication [23], and wireless sensor networks (wormhole attack [27]). Brands and Chaum have proposed distance bounding protocols [7] to mitigate these attacks (see [1] for further details).

3.2 External advisors

When computers started to be stronger than chess grandmasters [35], some players have seized this opportunity to cheat. Instead of using their brain on the board, they use it to elaborate strategies to receive computer advices during their games. This is strictly forbidden by the chess rules (see Appendix A) but the dishonest players have been extremely cunning as shown in the examples given in the paper introduction. We give three examples of scenario in which external advisors appears. The first one corresponds to the fraud's allegation 2010 chess Olympiad. We call the second and third examples respectively the *puppeteer fraud* and the *twin fraud*.

EXAMPLE 1 (THE FRENCH CONNECTION). *Let us imagine the following situation: Bob, an honest player challenges Alice, a dishonest one. Alice has two accomplices Charlie and Denis. Charlie is near Alice's board and can observe the game. Denis is far away running a chess software on his computer. Charlie observes Bob's move and transmits it to Denis. Denis obtains a proposal from the software for the best move and sends it back to Charlie. The last issue*

in the fraud is how Charlie communicates the move to Alice without attracting the attention of Bob. He can use a covert channel using body language or any other way that may not rise the suspicion of Bob. The scenario described above is supposed to be the one used by three French grandmasters during the 2010 chess Olympiad [32]. In the fraud allegation, Charlie used a cell phone to communicate with Denis. As cell phones are banned from the playing venue, Charlie needs frequently to go outside to send (receive) the moves to (from) Denis. The covert channel used by Alice and Charlie is detailed in Section 4.

EXAMPLE 2 (THE PUPPETEER FRAUD). *A chess grandmaster has been banned from chess tournaments. He hires two accomplices to continue to make money from chess. The first accomplice, i.e., the puppet, registers to the tournament while the grandmaster stays at home. The other accomplice is a simple spectator. In the playing venue, the accomplice observes the board and controls the puppet using a covert channel. Outside the playing venue, the accomplice communicates with the grandmaster using a cell phone. The fraud is described in Figure 1. At the end, they share the price.*

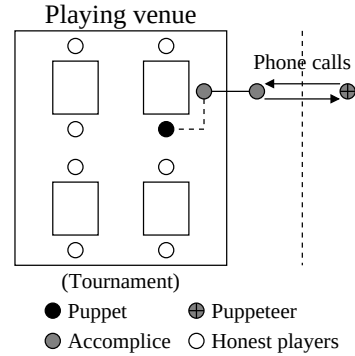


Figure 1: The puppeteer grandmaster. The puppeteer operates indirectly the puppet using one "strings": the accomplice in the public.

LINKS WITH COMPUTER SECURITY. Bengio, Brassard, Desmedt, Goutier and Quisquater [3] introduced a kind of man in the middle attack known as the *terrorist fraud* that matches our problem. Alice, a legitimate prover in an authentication protocol, collaborates with an accomplice Eve to impersonate herself to Bob. The motivations for this unexpected collaboration are various. Eve can rent Alice's identity. It corresponds to Example 1 in the chess world. Alice can pay Eve to help her to get an alibi from Bob. It corresponds to Example 2. More details, further references and countermeasures to the terrorist fraud are given in [2].

EXAMPLE 3 (THE TWIN FRAUD). *This fraud is easy to summarize: a chess game is started by one player and finished by another one exterior to the tournament. It requires naturally that the two switching players are colluding and they must be perfect twins to deceive the opponent and the arbiter. The benefit of such a strategy is that one of the*

twins can be an expert in the opening moves and the other a specialist of endings. To mount their attack, the twins need to find an opportunity to switch without being noticed.

LINKS WITH COMPUTER SECURITY. A variant of the distance hijacking attack proposed by Cremers, Bonne Rasmussen and Capkun [14] is also related to the problem of external advisors in chess. Without giving more details on [14], their attack against Brands and Chaum’s protocol [7] can be summarized as follows: *one phase of the protocol is executed by a prover and the final phase is executed by another different one.* This is exactly the twin fraud.

3.3 Players collusion

In collusion frauds, everything takes place at the playing venue: no outsider interferes with the course of a game or with the tournament result. This kind of fraud corresponds to the case where an agreement is made by some players. The agreement can be losing on purpose or providing advices to another player in the playing venue.

LOSING ON PURPOSE. A coalition of players agrees to make winning one of them. Depending on the course of the tournament and the identities of his adversaries, a member of the coalition may loose or draw on purpose. The malicious player plays his game normally until he makes a blunder. By this way, he does not attract the attention of the arbiter. The benefit of such an attack can be analyzed using the game theory. A famous story of collusion allegation occurs at the 5th Candidates Tournament at Curaçao in 1962 [46].

EXAMPLE 4 (CURAÇAO 1962). *In the middle of the cold war, five USSR grandmasters (Petrosian, Keres, Geller, Korchnoi and Tal), two Americans (Fischer and Benko) and a representative of Czechoslovakia (Filip) battle to get the right to challenge Mikhail Botvinnik the World Chess Champion. After the victory of USSR grandmaster Petrosian, three allegations of cheating were emitted. Fischer claimed that the USSR grandmasters Petrosian, Keres and Geller colluded to prevent him from finishing in the first three places (granting a direct access to the next cycle for the world title). He also claimed that Korchnoi lost on purpose against the previous three grandmasters in order to grant them the top places. Finally, it was said that Petrosian and Geller helped Benko in his game against Keres to grant Petrosian overall victory. These allegations have motivated the choice of the Fédération Internationale Des Échecs (FIDE) for changing the organization of the battle for the world title.*

LINKS WITH COMPUTER SECURITY. Collusion in computer security is also a major concern. In cryptography, for example, secret sharing [39] aims to avoid collusion among people. A typical use consists in sharing a highly sensitive secret, such that any predefined subset of people can retrieve it. It is used for example in e-voting applications, or by certification authorities to protect the access to their root certificate. In network security, collusion is usually addressed with game

theory-based mechanisms of reputation-based mechanisms. The problem occurs for example in routing protocols where rational nodes may collude to increase their personal bandwidth [50]. Losing on purpose can then be a local strategy to increase the global gain of the colluding parties.

INTERNAL ADVISORS. A method to implement a collusion fraud consists to provide advices to the other members of the coalition. The methods used, *i.e.*, covert channels, are exactly the same as in the case of external advisors. The only difference is that everything take place in the playing venue. An example of this strategy is given in what follows.

EXAMPLE 5 (LAST ROUND COLLUSION). *Let consider a tournament with four players, Alice, Bob, Charlie, and Denis. Before the last round, Alice is ahead of the others. She plays against Denis and if she wins or draws, she will be definitively victorious. Otherwise, the winner of the game between Bob and Charlie would earn the prize. At first sight, Denis has no hope of gain. In exchange of part of the prize, Denis offers to Bob and Charlie his agreement to be advised by them: he becomes their puppet. By helping Denis, Bob and Charlie increase the probability that Alice loses and therefore their own success probability. If they agree, the accomplices will plan a communication scheme to help Denis during his game against Alice.*

LINKS WITH COMPUTER SECURITY. Internal advisors are similar to external advisors and therefore connected to the terrorist fraud. However, the missing external adversary changes everything. The solutions found against the terrorist fraud can not be applied to the case of internal advisors.

4. COVERT CHANNELS

Covert channels [29, 33] are a well known problem in online games [34] or bridge [48]. During a chess tournament, covert channels are very easy to setup. First, the human body is the perfect medium: a game spectator can use a body language, *i.e.*, hands, eyes, mimic... to transmit information. The spectator can also use his moves in the playing venue. This later strategy was used at the chess Olympiad. But there are many more. For instance, many players (amongst which one of the authors), are used to play with the pieces they have taken during a game. This can be interpreted either as stress or as a way to exchange information with somebody. The same situation arises in poker games. Many poker players play with their buckets during a game intentionally or not. There are two interpretations: it can be either a stress sign which may help an opponent to make a decision or it is intended to another player. It renders the detection of these *visual covert channels* extremely hard.

A player or a spectator making many signs will become suspicious for the arbiters. Moreover, long communication requires to have the full attention of the cheaters: it is prone to error. The information must be represented efficiently to reduce the number of signs. This issue is discussed in this section. First, we recall the naive solution used by the French grandmasters and then suggest more efficient solutions.

THE FRENCH SCHEME. During the Olympiad, the French coach uses his movement in the playing venue to communicate with the player. The coach stops in front of a given chessboard. If he is by the side of the black or of the white, the player understands a different coordinate. The move representation associated to the coach position is the long algebraic chess notation: the rows of the board are identified by a number and the columns by a letter. The symbol (letter,number)-(letter,number) indicates the starting square and the destination square (see Figure 2). Such a scheme requires twelve bits to communicate a move. The coach and the player agrees on 64 positions in the playing area before the tournament. The coach needs to occupy two positions to indicate a move.

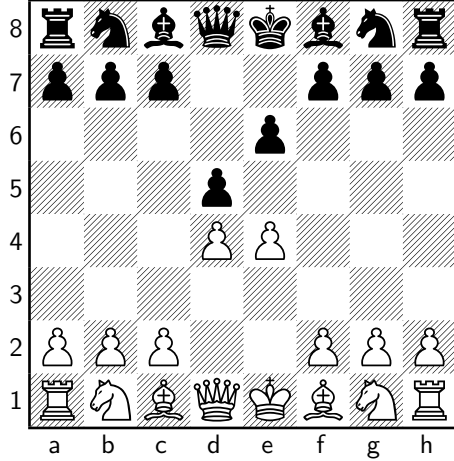


Figure 2: Long algebraic chess notation. Each move is described by the starting coordinate of the piece and by the destination for instance the figure corresponds to: 1.e2-e4 e7-e6 2.d2-d4 d7-d5. The starting square is often omitted in the algebraic notation if there is no ambiguity.

The French grandmasters used an inefficient representation of the board. We show how to use less bits.

BRANCHING NUMBER CODING. The compact communication of a chess move is fully answered by Shannon's works on communications [40] and chess [41]. The entropy, *i.e.*, the uncertainty, is for the chess player the branching number of the position, *i.e.*, the number of legal moves he can make on the board and denoted b_p . Shannon estimates in [41] that the maximum number of moves available to a chess player is in the order of $\max(b_p) \approx 35$. The player and the accomplice needs to agree on a method to enumerate all the moves (for instance from left (a) to right (h) and top (8) to bottom (1)). So, the accomplice needs to communicate $\lceil \log_2(b_p) \rceil$ bits to the player. Note there is no ambiguity with such a coding system, two pieces can reach the same case but these two different moves have different numbers. So in the worst case, we need six bits which is a big improvement over the twelve bits of the naive solution. An example is given in Figure 3 and Table 1 enumerates the possible moves. The accomplice recommends the move number 5 (three bits) to the player. With algebraic notations, the accomplice needs

to communicate **c5-d6** (twelve bits).

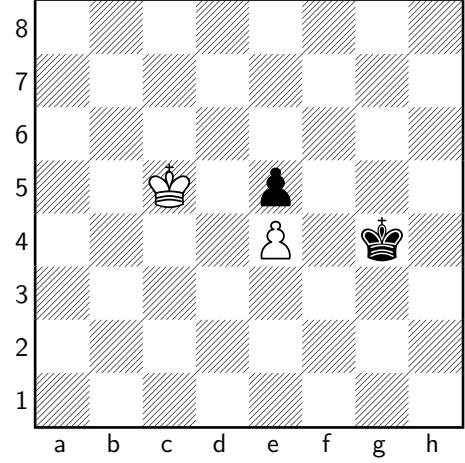


Figure 3: Seven moves are possible for the white.

Move	♔b6	♔b5	♔b4	♔c6	♔c4	♔d6	♔d5
Number	0	1	2	3	4	5	6

Table 1: Enumeration of the legal moves in Figure 3.

Additional source coding can be employed to further improve the conciseness but they must be easy to decode for the human brain (see [38, 45, 49]). The use of prefix codes [28] is possible but it is not discussed in this paper.

HUMAN-FRIENDLY CODING. If branching number coding is simple to decode in the opening or in the endgame, it is far more difficult in the middle game: the player can make errors when he enumerates the legal moves. A simpler solution consists in first sending a piece identifier and then a move identifier for the piece. The player obtains the bits from its accomplice using a covert channel. He computes the number of pieces which can make a legal move, it gives him the number of bits corresponding to the piece identifier. He makes the correspondence between the board and the piece identifier respecting an enumeration order pre-established with his accomplice (for instance left to right and top to bottom). The remaining bits identify the piece move. The player enumerates all the possible moves of the piece according to the pre-established order and move the piece accordingly. There is no need of separator between the piece identifier and the move identifier because the player can compute prior to the transmission the number of bits needed by the piece identifier.

In some case, this coding scheme is as efficient as branching number coding but far more easy to decode for the player. In the worse case, the accomplice needs to communicate nine bits to describe a queen move with 27 possible destinations (five bits) on a board in which all the other pieces can move (four bits). The Table 2 shows the maximum number of possibilities for each piece.

In Figure 4, the accomplice needs to send six bits to give

Piece						
Max. moves	8	27	14	13	8	4 (+4)

Table 2: Maximum number of legal moves for each piece (do not forget that pawn can be promoted!).

the black winning move. There are thirteen pieces which can make a legal move (four bits for the piece identifier) and the queen has four moves possible (two bits for the move identifier of the queen). The accomplice sends 5 3 to the player (2... ♖d8–h4♔). With branching number coding, he can save a bit: there are 30 moves legal for the black player.

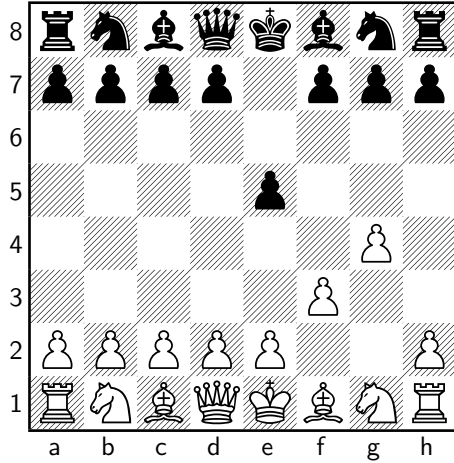


Figure 4: Mate in one!

REMARK 1. For comparison purpose, we assume that the player and his accomplice are using a binary covert channel. In practice, they are not restricted to a binary channel. They can exchange larger symbols using body language. In this way, they can improve the efficiency of the information transfer.

5. COUNTERMEASURES

Some proposals inspired by computer security can be made to ensure chess tournament fairness. They can be split in two categories: non-invasive and invasive countermeasures. The difference between these two kinds is that non invasive countermeasures do not affect the behavior of the player during the game, while invasive ones actually do. We discuss the countermeasures below, then study their consequences.

5.1 Non Invasive Countermeasures

DETECTOR/JAMMER. Detecting covert channels is a real problem in computer security. For instance, we would like to ensure in a network that there is no infected computer (by trojan horse, or bot...) leaking information to a remote hacker. However, detecting a covert channel might be hard, so a brute force approach consist in jamming all the channel. Jammers are forbidden in many countries. However, it is authorized in some countries (like France) under special

circumstances, *e.g.*, in theaters, to avoid cell phone ringing. In chess context, body search or X-ray allow to detect communication devices. It is pretty clear that covert channels using body language are harder to detect. A behavioral analysis of the players and their environment can solve a part of this issue but it is an expensive and hazardous solution. The wireless channel is easier to jam than a visual channel. Jamming visual covert channels seems to be particularly difficult to realize without breaking the players' concentration.

PARTIAL ISOLATION. The covert channels issue can be treated beforehand. This is what happened in sensitive environment relying on a computer infrastructure, *e.g.*, TEMPEST standards are used by NATO to circumvent covert channels. Partial isolation is also used in smartcard context to make them tamper resistant. In chess tournaments, it is impossible to have a playing area certified TEMPEST without modifying the players' behavior. Partial isolation is still possible. At the Sofia M-Tel tournaments [31], it has been used: the playing area was a *sound-proof glass cube*. The audience can observe the players without disturbing them. Moreover, isolating the players from the rest of the word prevents them from getting help from a computer. On one hand, the honest audience can act as an extra arbiter. On the other hand, accomplices in the audience can communicate with the players using visual covert channels. One-way glass solves this issue. In any case, partial isolation does not prevent players to receive help from other colluding players.

Jammers and partial isolation are certainly the only solution affordable to the majority of tournament organizers. However, they are not sufficient to prevent collusion between players.

5.2 Invasive Counter measures

TOURNAMENT MODE. Round-Robin tournaments (all-play-all tournaments) are easily controlled by coalition of malicious players, as suggested by the 5th Candidates Tournament at Curaçao in 1962. After this event, the FIDE decided to switch from Round-Robin tournament, all to elimination matches to determine the opponent to the World Chess Champion. However, elimination matches are less fair than the Round-Robin. What is gained in term of security is lost in fairness.

TIME CONTROL. As shown in Section 3, the attacks studied in this paper are not so different from those found against authentication protocols. So why not using solutions against those attacks in our context. *Distance bounding protocols* are a defense proposed by Brands and Chaum [7] against the mafia or terrorist fraud. These protocols combined authentication and distance upper bounding by timings (see [1]). Time is an important dimension in chess game (clock). Transposing distance bounding protocols to solve chess problems can be done using an *hour glass*: the player has not enough time to mount complex fraud and keeps his attention on the board. However, it reduces the possibilities offered by chess digital clock. In rapid chess game, *e.g.*, blitz, getting external help is clearly more difficult than in for slower

games [18].

In a first attempt to secure chess at the game level, we propose Jail tournament. It has two goals: preventing help from outside the playing venue and eradicating collusion. The first objective can be achieved by playing tournaments in a jail. The second objective is more difficult to solve because we have to deal with the well-known *prisoners problem*. Applying it on chess allow malicious players to use the chess board itself as a covert communication channel. To defeat this last challenge, we introduce Fischer random chess Jail tournament.

JAIL. We aim at totally isolating the players from the rest of the world. The best solution consists in putting all the players in individual cells in which they have no way to communicate with the outside or with another cell. The players are not allowed to leave this “confinement cell”. They can only play chess and have some basic commodities during the tournament. All the games are remotely controlled by the arbiter (see Figure 5).

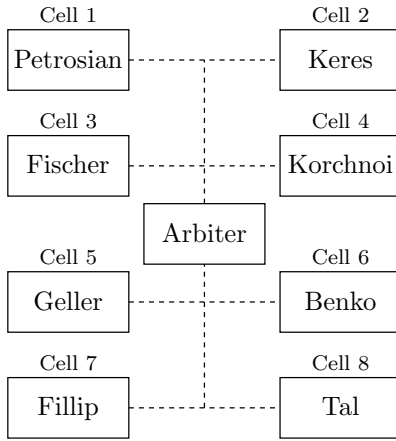


Figure 5: The setup of a Jail tournament. Each player is kept into a cell isolated from the rest of the world. Each cell contains a screen, a mouse and all the commodities needed for the duration of the tournament. The screen displays the chessboard of the player game and the game clock (his adversary time and his own time left to complete the game). The screen displays no information on the identity of the adversary. All the screens and mice are controlled by the arbiter. He decides the pairing between the players and he is the only one knowing them. A side effect of this setup is that we can strongly enforce “touch-move” rules.

ANONYMOUS PAIRING. The arbiter is the only one knowing the exact identities of the players of a game. We assume that the pairing method is the Round-Robin. The purpose of this policy is to prevent collusion between the players. The dishonest players must not be able to determine when they are facing an accomplice. The arbiter needs to eradicate any communication scheme which can help the players to

identify themselves. This is particularly difficult since it is related to the prisoners problem.

PRISONERS PROBLEM. This problem was introduced by Simmons [42, 43, 44]. It describes the situation of two accomplices who have committed a crime, and have been caught. Now, they are locked in separate cells. They want to escape, so they need to agree on details. However, the only way for them to communicate is through the warden. The latter wants to avoid the escape. So, the two accomplices need to find a subliminal channel to discuss the details of their escape without revealing anything about their plan to the warden. Another condition is that the warden forbids them to encrypt exchanged messages.

With jail tournaments, a coalition of dishonest players is in the same situation than the two prisoners. They need to communicate with their opponent (who is the only one they can currently reach) to see whether he is part of the coalition. In 1944, the Allies face the same problem. They used “airborne cricket” to identify themselves without revealing their presence to the Germans. To defeat anonymous pairing, the dishonest players can use a similar trick. The coalition members agree on special moves orders prior going into their cells. These moves orders are then used during the tournament to signal the coalition membership. The dishonest players must be discreet because the arbiter can eavesdrop easily the channel they use. The principle of this method is illustrated in the following example.

EXAMPLE 6 (SUBLIMINAL IDENTIFICATION). *We demonstrate that it is relatively easy to find a way for the coalition members to identify themselves over the board. The position described in Figure 6 can be reached by four different move orders:*

- (a) 1 e4 c5 2 ♠f3 d6,
- (b) 1 e4 d6 2 ♠f3 c5,
- (c) 1 ♠f3 c5 2 e4 d6,
- (d) 1 ♠f3 d6 2 e4 c5.

A database exploration of more than four millions games [12] shows that 273247 parties reached the position of Figure 6: 99.45% used move order (a), 0.03% for (b), 0.45% for (c) and 0.07% for (d).

A coalition of players can decide that they all start their games with ♠f3. If they reach the position of Figure 6 using move order (c) or (d) they conclude that they are playing against a member of the coalition. If all the players used the same move order to reach a position, the arbiter might be suspicious. Each player of the coalition can be assigned a set of moves orders used only by him. This set will act as a signature to all the other members of the coalition. At the end, the arbiter can only conclude that the players have decided to use unusual move orders.

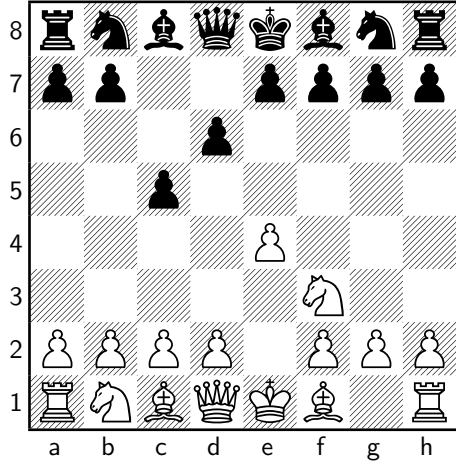


Figure 6: A variant of the Sicilian opening.

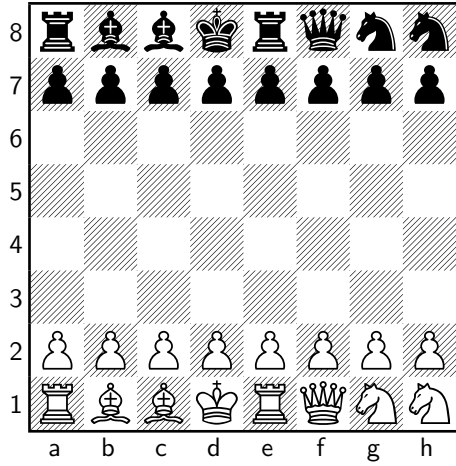


Figure 7: A starting position at FRC (SPID 147).

FISHER RANDOM CHESS. Anonymous pairing is not sufficient to remove collusion because the players can communicate through the board. To remove this communication channel, a solution can be to use randomization as advocated in [26]. Randomization is possible if we allow a slight change in the chess rules. Instead of starting all the games with the same position, the initial position is chosen randomly. This variant of chess is known as Fisher random chess or chess960 or Fischerandom Chess, and it is detailed in Appendix B. The positions of the pawns are the same as in classical chess and the differences are the positions of the other pieces and the way of castling (Figure 7). There are 960 different starts which is not impressive for a computer but already difficult to handle for a human brain.

REMARK 2 (TIMING CHANNELS). *Fisher random chess can mitigate subliminal channels over the chess board. However, timing channels are also possible: players see when their opponents play their moves. Timing channels are risky. Dishonest players going too quickly or too slowly may have a disadvantage against fair players. Moreover, randomiza-*

tion can be used. Let consider that a game in which each player has $\ell + k$ minutes to complete his game. The value ℓ is the fixed amount of time distributed to each player initially. The value k is an increment: extra time given to the players. At each move, the player receives a fraction m of these k minutes. If m is a constant value, timing channel is still possible. The principle of this time control has been proposed by Fischer and is very popular nowadays in international competition. If the arbiter choses randomly the value m for each move timing channels can be mitigated exactly as in [26]. To enforce this solution, the arbiter must remove any clock in the players cell.

5.3 Consequences

To conclude this section, Table 3 resumes the benefit of the different isolation levels in a chess tournament. Each of these levels aims at avoiding a particular fraud. Isolating the player in a jail, *i.e.*, at the playing venue level, prevents from external advisors. Isolating the player from the other ones, *i.e.*, at the game level, prevents from insider advisors. Finally, isolating the player from his opponent, *i.e.*, at the player level, prevents from collusion between the players with the help of anonymity. Transposing these methods to authentication, we can say that distance bounding protocols are a method to defeat external threats by ensuring that the prover is in a trusted neighborhood, *i.e.*, the playing area. Inside this neighborhood, distance bounding protocols assume that no attack can occur. Actually, there is no detail on how can be implemented the trusted neighborhood. Studying chess problems shows us that isolation is a solution to implement the neighborhood concept in distance bounding protocols.

Isolation level	Prevent
Playing venue	external advices
Game	collusion (internal advices)
Player	collusion (coalition)

Table 3: Benefits of the different isolation levels.

REMARK 3. *An interesting question is to know how fair it is to have the identity of your opponent. Indeed, the opponent identity influences the players strategy. Our modification of chess tournaments tends to remove the psychological side of chess to keep only the pure chess strength.*

CHESS PLAYERS PRIVACY. The previous remark and the problem of subliminal identification have highlighted the question of the player's identity. If my opponent does not know my identity can he guess who I am. This is particularly interesting for online chess. All chess players have their pet openings and some move orders they favor in specific positions. Using a database [12], a malicious player can compute the profile of all players in the database. Then, he can try to find a match between the profiles and the statistics collected from the games played online against anonymous adversaries. In other words, *show me how you play and I will tell you who you are!* Figure 8 shows the profile of five top grandmasters (Anand, Carlsen, Aronian, Kramnik and Ivanchuk) when they play their first move with the white.

The number of games analyzed for each player is between 700 and 1500. The Figure 8 shows the characteristics of each grandmaster.

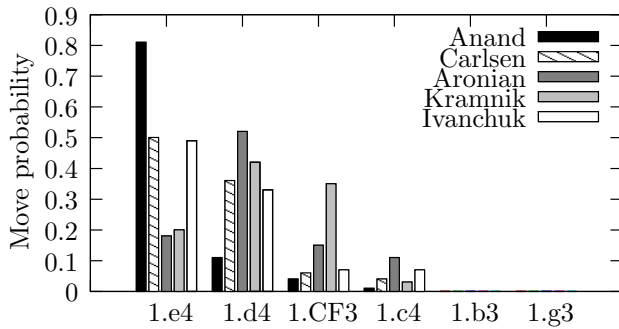


Figure 8: Opening profile for five top grandmasters.

6. CONCLUSION

Grandmaster Gligoric advocates in his book *Shall We Play Fischerandom Chess?* [24] for the end of the classic chess era to the benefit of FRC. His main argument is that computers plague classical chess even in fair games: players make deep training on chess openings, rendering games boring. He points out that FRC makes more room for creativity and reduces the impact of computerized training.

Our paper also goes into that direction, and deeper, disclosing several chess frauds – theoretical or already exploited – and provides some countermeasures to avoid or mitigate them. We mostly advocate for two radical transformations to ensure the fairness of chess tournaments. First, chess players must be isolated in individual cells without knowing the identity of their adversary (Jail tournament). Second, we consider, as Gligoric suggested, that classical chess should be abandoned to favor Fischer random chess.

Our work is mostly based on a comparison between chess and computer security, in order to identify how frauds can be mounted in chess tournaments, but also to find practical solutions. Not all our countermeasures can be easily put in place in face-to-face tournaments, but we think they should be considered in the design of online chess tournaments. Beyond chess, frauds and countermeasures also apply to other board games like Shogi, Go, and even Poker.

We think this work raise interesting questions that may be thoroughly addressed in the future. For example, time-based or opening-based covert channels are not desirable features. They can be used to exchanged covert information, but they may also reveal the identity of a player in anonymous on-line tournaments. We demonstrated that avoiding these two covert channels is far from being trivial. Techniques from computer security should be explored in order to find solutions. For example, techniques to protect smartcards against timing-attacks could be used to mitigate time-based covert channel; techniques used in Intrusion Detection Systems (IDS) to detect abnormal behaviors could also be used to avoid opening-based covert channel.

Finally, avoiding frauds in chess can be done by mutually

isolating chess players and security experts, but we consider that a better approach is to join their efforts to fight chess frauds. That is why we recommend chess players to learn computer security.

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7. REFERENCES

- [1] G. Avoine, M. A. Bingöl, S. Kardas, C. Lauradoux, and B. Martin. A Framework for Analyzing RFID Distance Bounding Protocols. *Journal of Computer Security*, 19(2):289–317, 2011.
- [2] G. Avoine, C. Lauradoux, and B. Martin. How Secret-sharing can Defeat Terrorist Fraud. In *ACM Conference on Wireless Network Security - WISEC 2011*, page To appear, Hamburg, Germany, June 2011. ACM.
- [3] S. Bengio, G. Brassard, Y. Desmedt, C. Goutier, and J.-J. Quisquater. Secure Implementation of Identification Systems. *Journal of Cryptology*, 4(3):175–183, 1991.
- [4] T. Beth and Y. Desmedt. Identification Tokens - or: Solving the Chess Grandmaster Problem. In *CRYPTO*, volume 537 of *Lecture Notes in Computer Science*, pages 169–177, Santa Barbara, CA, USA, August 1990. Springer-Verlag.
- [5] J. Black, M. Cochran, and M. R. Gardner. Lessons Learned: A Security Analysis of the Internet Chess Club. In *Annual Computer Security Applications Conference - ACSAC 2005*, pages 245–253, Tucson, AZ, USA, 2005. IEEE Computer Society.
- [6] J. Black, M. Cochran, and M. R. Gardner. A Security Analysis of the Internet Chess Club. *IEEE Security & Privacy*, 4(1):46–52, 2006.
- [7] S. Brands and D. Chaum. Distance-Bounding Protocols. In *Advances in Cryptology - EUROCRYPT’93*, volume 765 of *Lecture Notes in Computer Science*, pages 344–359, Lofthus, Norway, May 1993. Springer-Verlag.
- [8] B. Byfield. *Every Great Chess Player Was Once a Beginner*. Lyle Stuart, 1978.
- [9] M. Chandler. *How to Beat Your Dad at Chess*. Gambit Chess, 1998.
- [10] ChessBase. Cheating at the World Open – more details , 2006. <http://www.chessbase.com/newsdetail.asp?newsid=3292>.
- [11] ChessBase. Topalov threatens to abandon the world championship match, 2006. <http://www.chessbase.com/newsdetail.asp?newsid=3370>.
- [12] ChessBase. Big Database 2011 , 2011. <http://www.chessbase-shop.com/en/products/5852>.
- [13] J. H. Conway. *On Numbers and Games*. Number 6 in London Mathematical Society Monographs. Academic Press, London-New-San Francisco, 1976.
- [14] C. Cremers, K. B. Rasmussen, and S. Capkun. Distance Hijacking Attacks on Distance Bounding

- Protocols. Cryptology ePrint Archive, Report 2011/129, 2011.
- [15] Y. Desmedt, C. Goutier, and S. Bengio. Special Uses and Abuses of the Fiat-Shamir Passport Protocol. In C. Pomerance, editor, *Advances in Cryptology – CRYPTO’87*, volume 293 of *Lecture Notes in Computer Science*, pages 21–39, Santa Barbara, CA, USA, August 1988. Springer-Verlag.
 - [16] S. Drimer and S. J. Murdoch. Keep your enemies close: distance bounding against smartcard relay attacks. In *USENIX Security Symposium on USENIX Security Symposium*, pages 1–16, Boston, MA, USA, 2007. USENIX Association.
 - [17] Europe Echecs. Triche présumée aux Olympiades, 2011. <http://www.europe-echecs.com/actualites/compte-rendu-de-la-commission-de-discipline-2583.html> (in french).
 - [18] Fédération Française des Échecs. Interview of Kasparov, 2011. <http://www.youtube.com/watch?v=-PQyDN19moY>.
 - [19] Fédération Internationale des Échecs. FIDE Laws of Chess, 2009. www.fide.com/FIDE/handbook/LawsOfChess.pdf.
 - [20] U. Feige, A. Fiat, and A. Shamir. Zero Knowledge Proofs of Identity. In *Proceedings of the Nineteenth Annual ACM Symposium on Theory of Computing – STOC*, pages 210–217, New York City, NY, USA, May 1987. ACM.
 - [21] A. Fiat and A. Shamir. How to prove yourself: Practical solutions to identification and signature problems. In *Advances in Cryptology – CRYPTO’86*, Lecture Notes in Computer Science 263, pages 186–194, Santa Barbara, CA, USA, August 1986. Springer-Verlag.
 - [22] B. Fischer, S. Margulies, and D. Mosenfelder. *Bobby Fischer Teaches Chess*. Bantam, 1982.
 - [23] A. Francillon, B. Danev, and S. Capkun. Relay Attacks on Passive Keyless Entry and Start Systems in Modern Cars. In *Network and Distributed System Security Symposium*, San Diego, CA, USA, February 2011. Internet Society.
 - [24] S. Gligoric. *Shall We Play Fischerandom Chess?* Batsford, 2003.
 - [25] G. P. Hancke. Practical Attacks on Proximity Identification Systems. In *IEEE Symposium on Security and Privacy - S&P 2006*, pages 328–333, Berkeley, CA, USA, 2006. IEEE Computer Society.
 - [26] W.-M. Hu. Reducing Timing Channels with Fuzzy Time. In *IEEE Symposium on Security and Privacy*, pages 8–20, Oakland, CA, USA, 1991. IEEE.
 - [27] Y.-C. Hu, A. Perrig, and D. B. Johnson. Packet Leashes: A Defense against Wormhole Attacks in Wireless Networks. In *INFOCOM*, San Francisco, CA, USA, March 2003. IEEE.
 - [28] D. A. Huffman. A Method for the Construction of Minimum-Redundancy Codes. *Proceedings of the IRE*, 40(9):1098–1101, 1952.
 - [29] B. W. Lampson. A note on the confinement problem. *Communication of the ACM*, 16:613–615, October 1973.
 - [30] D. L. McClain. CHESS; Cheating Accusations in Mental Sports, Too. *New York Times*, August 2006. <http://query.nytimes.com/gst/fullpage.html?res=9502E3D71E3FF93BA3575BC0A9609C8B63&pagewanted=1>.
 - [31] D. L. McClain. Carlsen Takes Early Lead at Grand Slam Masters. *New York Times*, September 2008. <http://gambit.blogs.nytimes.com/2008/09/03/carlsen-takes-early-lead-in-bilbao/>.
 - [32] D. L. McClain. Three French Players Punished for Using Technology to Cheat. *New York Times*, April 2011. <http://www.nytimes.com/2011/04/03/crosswords/chess/03chess.html>.
 - [33] J. K. Millen. 20 Years of Covert Channel Modeling and Analysis. In *IEEE Symposium on Security and Privacy*, pages 113–114, Oakland, CA, USA, May 1999. IEEE Computer Society.
 - [34] S. J. Murdoch and P. Zielinski. Covert channels for collusion in online computer games. In *International Workshop on Information Hiding - IH 2004*, Lecture Notes in Computer Science 3200, pages 355–370, Toronto, Canada, May 2004. Springer-Verlag.
 - [35] M. Newborn. *Kasparov Versus Deep Blue: Computer Chess Comes of Age*. Springer-Verlag, 1997.
 - [36] A. Newell, J. Shaw, and H. Simon. Chess-Playing Programs and the Problem of Complexity. *IBM Journal Research and Development*, 2(4), 1958.
 - [37] Phonak Communications AG. Phonito Nano, 2006. <http://www.phonak-communications.com/en/security/downloads/>.
 - [38] S. S. Pradhan and K. Ramchandran. Distributed source coding using syndromes (DISCUS): design and construction. *IEEE Transactions on Information Theory*, 49(3):626–643, Mars 2003.
 - [39] A. Shamir. How to share a secret. *Commun. ACM*, 22(11):612–613, 1979.
 - [40] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 1948.
 - [41] C. E. Shannon. Programming a computer for playing chess. *Philosophical Magazine* (7), 41(314):256–275, March 1950.
 - [42] G. J. Simmons. The Prisoners’ Problem and the Subliminal Channel. In *Advances in Cryptology – CRYPTO ’83*, pages 51–67. Plenum Press, 1983.
 - [43] G. J. Simmons. A Secure Subliminal Channel (?). In *Advances in Cryptology - CRYPTO ’85*, Lecture Notes in Computer Science 218, pages 33–41, Santa Barbara, CA, USA, August 1985. Springer-Verlag.
 - [44] G. J. Simmons. The History of Subliminal Channels. In *First International Workshop on Information Hiding*, Lecture Notes in Computer Science 1174, pages 237–256, Cambridge, U.K., May 1996. Springer-Verlag.
 - [45] D. Slepian and J. K. Wolf. Noiseless coding of correlated information sources. *IEEE Transactions on Information Theory*, 19(4):471–480, Jul 1973.
 - [46] J. Timman. *Curaçao 1962, the Battle of Minds that shook the International Chess World*. New in Chess, 2005.
 - [47] J. von Neumann. Zur Theorie der Gesellschaftsspiele. *Mathematische Annalen*, 100(1):295–320, 1928.

- [48] P. Winkler. The Advent Of Cryptology In The Game Of Bridge. *Cryptologia*, 7(4):327–332, 1983.
- [49] A. Wyner. Recent Results in the Shannon Theory. *IEEE Transactions on Information Theory*, 20(1):2–10, Mar 1974.
- [50] S. Zhong and F. Wu. On designing collusion-resistant routing schemes for non-cooperative wireless ad hoc networks. In *MOBICOM*, pages 278–289, Montréal, Canada, 2007.

APPENDIX

A. RULES OF CHESS

We provide some relevant extracts of Article 12 and 13 of the FIDE Laws of Chess [19]. Article 12 defines the conduct of the players and Article 13 the role of the arbiter.

A.1 Extracts from article 12

The interesting part in the FIDE rules starts with Article 12 related to the conduct of the players. The point 12.2 defines where a chess player can move during an official game.

- 12.2 Players are not allowed to leave the “playing venue” without permission from the arbiter. The playing venue is defined as the playing area, rest rooms, refreshment area, area set aside for smoking and other places as designated by the arbiter. The player having the move is not allowed to leave the playing area without permission of the arbiter.
- 12.3 a. During play the players are forbidden to make use of any notes, sources of information or advice, or analyse on another chessboard.
- b. Without the permission of the arbiter a player is forbidden to have a mobile phone or other electronic means of communication in the playing venue, unless they are completely switched off. If any such device produces a sound, the player shall lose the game. The opponent shall win. However, if the opponent cannot win the game by any series of legal moves, his score shall be a draw.
- c. Smoking is permitted only in the section of the venue designated by the arbiter .

A.2 Extracts from article 13

- 13.1 The arbiter shall see that the Laws of Chess are strictly observed.
- 13.7 a. Spectators and players in other games are not to speak about or otherwise interfere in a game. If necessary, the arbiter may expel offenders from the playing venue. If someone observes an irregularity, he may inform only the arbiter.
- b. Unless authorised by the arbiter, it is forbidden for anybody to use a mobile phone or any kind of communication device in the playing venue and any contiguous area designated by the arbiter.

B. FISCHER RANDOM CHESS

Fischer random chess was introduced by Fischer in 1996 based on an existing chess variant called Shuffle Chess. The rules of Fischer random chess are essentially the same except for the starting position and the castling.

The starting position is chosen randomly amongst 960 possible positions. For both sides, the initial pawns position is the same than in classical chess. The position of the other pieces is chosen randomly with two notable restrictions:

- the king must be placed between the two rooks,
- the bishops must be placed on opposite-colored squares.

The black pieces are equal-and-opposite to the white ones, e.g if the white king is placed in b1, the black one has to be on the b8. The Starting Position Identifier (SPID) designs each of the 960 possible initial positions. SPID 518 corresponds to the starting position of classical chess.